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UVC LED Buck Converter Driver for Water Disinfection

Project Summary

A closed-loop asynchronous buck converter driving an 8-LED Crystal IS Klaran UVC array for *E. coli* inactivation in water, designed and simulated in LTspice, with PI current control firmware on an Arduino controller.

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This document describes the design, simulation, firmware development, and hardware prototyping of a UVC LED driver for water disinfection research.

Contents

1. Introduction

Waterborne diseases caused by *Escherichia coli* and other pathogens remain a significant global health challenge. Ultraviolet-C (UVC) radiation in the 260 nm to 270 nm germicidal range disrupts bacterial DNA by forming pyrimidine dimers, preventing replication. A UV dose of 15 mJ/cm^2 to 20 mJ/cm^2 at 265 nm achieves a 4-log (99.99 %) reduction of *E. coli* populations [1].

Solid-state UVC LEDs offer significant advantages over traditional low-pressure mercury lamps: instant-on capability, no hazardous mercury, compact form factor, and the ability to target specific germicidal wavelengths. However, they require precision constant-current drivers to maintain stable optical output and junction temperature within safe operating limits.

This report describes the complete design flow of a UVC LED buck converter driver: from specification and component selection, through LTspice simulation, Arduino firmware development, breadboard prototyping and PCB design, targeting the Crystal IS Klaran SA265-35030-SA1 UVC LED array.

2. System Specifications

2.1. Target LED – Crystal IS Klaran SA265-35030-SA1

The key electrical and optical parameters of the selected UVC LED are summarized below.

Table 1: Klaran SA265-35030-SA1 Key Parameters

Parameter	Symbol	Min	Typical	Max
Peak wavelength at 500 mA	λ_P	260 nm	–	270 nm
Forward voltage at 500 mA	V_F	5.0 V	–	9.0 V
Optical power at 350 mA	P_t	30 mW	–	–
Thermal resistance (junction-to-solder)	$R_{\theta JS}$	–	7 °C/W	–
Max junction temperature	T_J	–	–	115 °C
Continuous forward current	I_F	100 mA	–	700 mA

2.2. LED Array Configuration

Eight LEDs are arranged in a **2-series** × **4-parallel** (2S4P) configuration, chosen to balance string voltage against available supply headroom:

$$V_{\text{string}} = 2 \times V_F = 2 \times 7\,\text{V} = 14\,\text{V} \quad \text{(typical)} \tag{1}$$

$$I_{\text{total}} = 4 \times 500\,\text{mA} = 2\,\text{A} \tag{2}$$

$$P_{\text{optical}} \geq 8 \times 30\,\text{mW} = 240\,\text{mW} \tag{3}$$

2.3. Converter Specifications

The target design requirements for the buck converter operating in continuous conduction mode (CCM) are detailed below.

Table 2: Buck Converter Design Specifications

Parameter	Symbol	Value
Input voltage	V_{in}	18 V
Output voltage	V_o	14 V
Output current	I_o	2 A
Switching frequency	f_s	20 kHz
Duty cycle (M1)	D	77.8 %
Inductor ripple	ΔI_L	155.6 mA (7.8%)
Output voltage ripple	ΔV_o	≤ 0.35 V
Conduction mode	–	CCM (all cases)

3. Circuit Design

3.1. Buck Converter Topology

An asynchronous buck converter topology was selected, using an N-channel power MOSFET (IRLZ44N) as the main switch and a Schottky diode (1N5822) as the freewheeling element. The asynchronous approach was chosen for simplicity in the prototype phase; synchronous rectification may be considered in future iterations for improved efficiency.

The operating duty cycle is derived from the volt-second balance principle:

$$D = \frac{V_o}{V_{\text{in}}} = \frac{14}{18} = 0.778 \quad (77.8\%) \tag{4}$$

3.2. Inductor Selection

The critical inductance requirement for CCM operation is:

$$L_{\text{crit}} = \frac{(V_{\text{in}} - V_o)\, D}{2\, I_o\, f_s} = \frac{4 \times 0.778}{2 \times 2 \times 20{,}000} = 38.9\,\mu\text{H} \tag{5}$$

The inductor value for a target ripple current of $\Delta I_L = 10\%$ of $I_o = 2$ A:

$$L = \frac{(V_{\text{in}} - V_o)\, D}{f_s\, \Delta I_L} = \frac{4 \times 0.778}{20{,}000 \times 0.2} = 778\,\mu\text{H} \approx 1\,\text{mH} \tag{6}$$

Verification of actual ripple with $L = 1$ mH:

$$\Delta I_L = \frac{(V_{\text{in}} - V_o)\, D}{L\, f_s} = \frac{4 \times 0.778}{0.001 \times 20{,}000} = 155.6\,\text{mA} \quad (7.8\%) \tag{7}$$

Since $L = 1\,\text{mH} \gg L_{\text{crit}} = 38.9\,\mu\text{H}$, CCM is confirmed across all operating cases ($V_o = 10\,\text{V}$ to $14\,\text{V}$).

3.3. Output Capacitor Selection

Using the correct CCM ripple formula:

$$C = \frac{V_{\text{in}}\, D\,(1-D)}{8\,L\,f_s^2\,\Delta V_o} = \frac{18 \times 0.778 \times 0.222}{8 \times 10^{-3} \times (20{,}000)^2 \times 0.35} = 0.62\,\mu\text{F} \tag{8}$$

A value of $C_2 = 10\,\mu\text{F}$ was selected, providing substantial margin above the minimum requirement and accounting for capacitor derating under DC bias.

3.4. Gate Drive Circuit & Bootstrap Transient Analysis

The IRLZ44N requires adequate V_{GS} enhancement for low $R_{DS(\text{on})}$. Because M1 operates as a high-side switch, its source voltage swings up to $V_{\text{in}} = 18\,\text{V}$ during conduction, necessitating a floating high-side gate driver.

In initial bench testing of a discrete bootstrap gate driver circuit, the node switching behavior was captured on an oscilloscope operating at $f_s = 19.98\,\text{kHz}$ (Figure 1). The captured waveform revealed severe transient ringing and voltage overshoot/undershoot at the switching transitions:

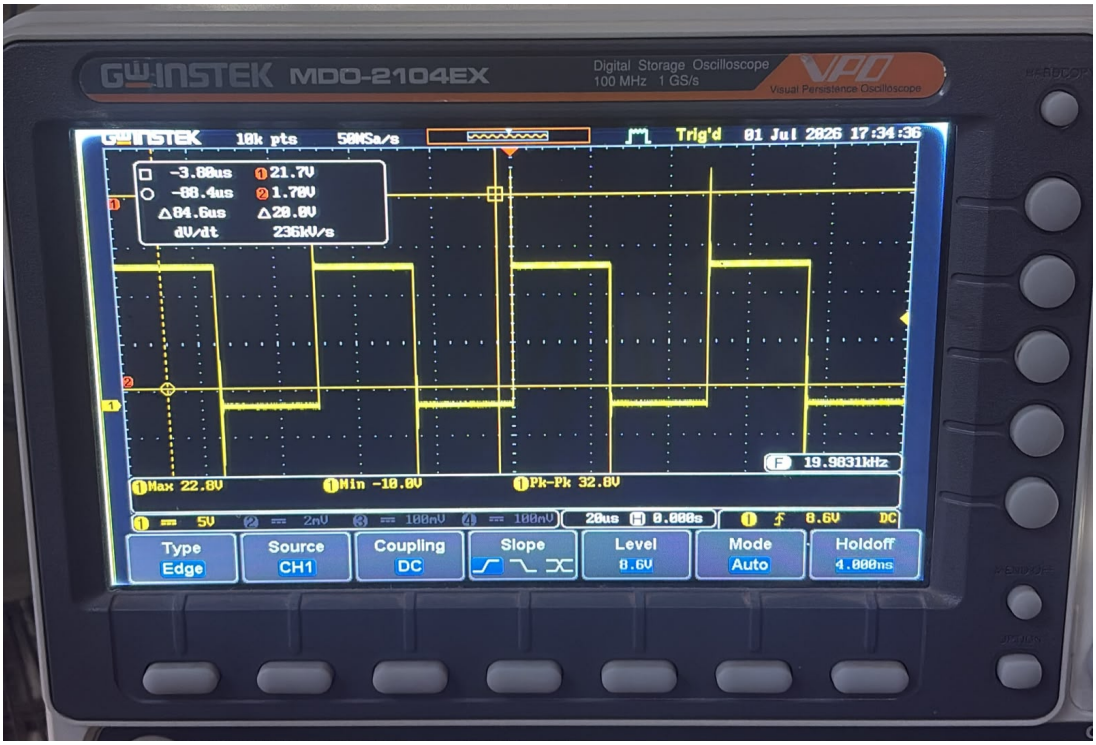

Figure 1: Oscilloscope trace of the gate switching node operating at 19.98 kHz, showing severe voltage overshoot spikes up to 22.8 V and negative undershoot down to −10.0 V ($V_{\text{pk-pk}} = 32.8\,\text{V}$) during discrete high-side switching.

As seen in Figure 1, high parasitic loop inductances and rapid $\text{d}V/\text{d}t$ transitions ($\approx 236\,\text{kV/s}$) produced peak-to-peak voltage excursions of $\Delta V_{\text{pk-pk}} = 32.8\,\text{V}$, with maximum

voltage spikes reaching +22.8 V and negative ringing undershooting to −10.0 V. These severe spikes risk exceeding the maximum V_{GS} limits of the MOSFET and introduce high levels of electromagnetic interference (EMI).

To mitigate these transients and ensure clean, fast switching, a dedicated IR2110 high-side gate driver IC combined with an RCD snubber network was integrated:

- **Gate driver IC U2** (IR2110): provides a non-inverting, low-impedance push-pull high-side gate drive (HIN to HO) driven directly from a 5 V Arduino PWM signal.
- **Bootstrap diode D3** (1N4148): charges the bootstrap capacitor C1 to approximately $V_{\text{in}} - V_f = 18 - 0.7\,\text{V} = 17.3\,\text{V}$ during the MOSFET OFF period.
- **Bootstrap capacitor C1** (10 µF): selected such that $C_1 \gg Q_g/\Delta V_{\text{gate}}$ where $Q_g = 83\,\text{nC}$ for IRLZ44N.

$$C_1 \gg \frac{Q_g}{\Delta V} = \frac{83\,\text{nC}}{1\,\text{V}} = 83\,\text{nF} \quad \Rightarrow \quad C_1 = 10\,\mu\text{F} \tag{9}$$

- **Gate resistor R1** (100 Ω): limits peak gate current and damps high-frequency ringing.

3.5. Snubber Circuit

An RCD snubber clamp was placed across the freewheeling diode D1 to suppress voltage spikes at the switching node. The network utilizes a fast-switching diode D2 (1N4148) in parallel with a bleed resistor R2 (12 kΩ) and in series with a capacitor C9 (30 nF).

During the main switch turn-on, D2 forward-biases, allowing C9 to quickly absorb high-frequency transient energy, which effectively clamps the ringing voltage. The capacitor then slowly discharges its stored energy through R2. The time constant of this discharge is:

$$\tau_{\text{snubber}} = R_2 \times C_9 = 12{,}000\,\Omega \times 30\,\text{nF} = 360\,\mu\text{s} \tag{10}$$

At a switching frequency of 20 kHz, the switching period is $T = 50\,\mu\text{s}$. Because the snubber time constant is significantly larger than the switching period ($\tau = 7.2 \times T$), the capacitor does not fully discharge between cycles. Instead, it acts as a stable DC voltage clamp that tracks the transient peak.

The continuous static power dissipation in the bleed resistor is safely clamped:

$$P_{R2} \approx \frac{V_{\text{in}}^2}{R_2} = \frac{(18\,\text{V})^2}{12{,}000\,\Omega} = 27\,\text{mW} \tag{11}$$

This confirms the chosen components operate well within the thermal limits of a standard quarter-watt (250 mW) resistor.

4. LTspice Simulation

4.1. Simulation Setup

Transient simulation was conducted in LTspice using the following control directive:

```
.tran 0 30m 0 100n
```

For full-power validation, the 8-LED UVC array was modeled by its equivalent full-load resistive load $R_3 = 7\,\Omega$ ($R_{\text{eq}} = V_o/I_o = 14\,\text{V}/2\,\text{A} = 7\,\Omega$).

To accurately capture high-side N-channel MOSFET driving dynamic behavior, the simulation incorporates a discrete driver stage with bootstrap acceleration. The PWM generator V2 driving the gate stage was configured as:

```
PULSE(0 5 0 10n 10n 11.1u 50u)
```

This directive sets a switching period of $T_s = 50\,\mu\text{s}$ ($f_s = 20\,\text{kHz}$) and an active duty cycle of $D = 0.778$ (77.8 %). In the LT spice image , IRFZ44N and 1N5818 have been used instead of IRLZ44N and 1N5822 as they do the same job and can be used interchangeably in the LT spice environment.

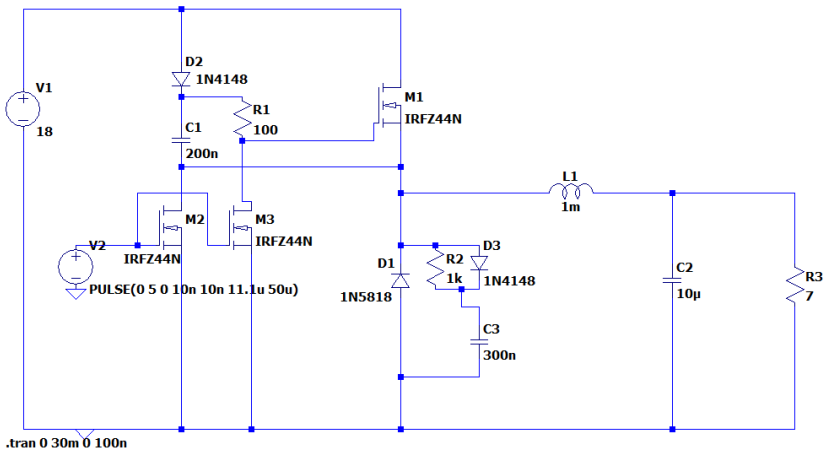

Figure 2: LTspice schematic of the asynchronous buck converter featuring the push-pull bootstrap gate drive circuit, snubber network, and a 7 Ω full-load resistor.

4.2. Simulation Results

The transient simulation results confirm steady-state performance against the initial design targets.

Table 3: LTspice Simulation Results vs Design Targets (Full Load)

Parameter	**Target**	**Simulated**	**Status**
Output voltage V_o	14.00 V	13.91 V (mean)	✓
Load current I_o	2.00 A	1.99 A (mean)	✓
Inductor ripple ΔI_L	≤ 200 mA	157 mA	✓
Output ripple ΔV_o	≤ 0.35 V	0.10 V	✓
Conduction mode	CCM	CCM confirmed	✓

As shown in Figure 3, the steady-state inductor current $I(L1)$ oscillates between 1.905 A and 2.062 A, giving a mean load current of 1.99 A and a peak-to-peak current ripple of:

$$\Delta I_L = 2.062 - 1.905 = 157\,\text{mA} \tag{12}$$

This closely aligns with the analytical target of 155.6 mA.

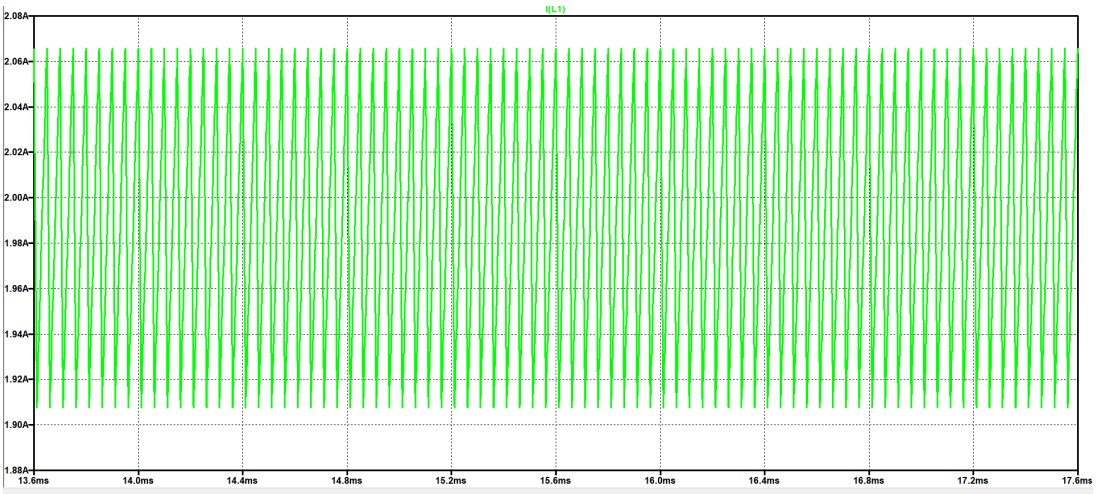

Figure 3: Simulated steady-state inductor current waveform (I_{L1}), demonstrating stable CCM operation with a mean current of approximately 1.99 A and a peak-to-peak ripple of 157 mA.

The steady-state output voltage waveform $V(n006)$ in Figure 4 settles at a mean of 13.91 V with peak-to-peak ripple bounds between 13.865 V and 13.965 V:

$$\Delta V_o = 13.965 - 13.865 = 100\,\text{mV} \quad (0.10\,\text{V}) \tag{13}$$

This minimal output ripple easily satisfies the design specification (≤ 0.35 V).

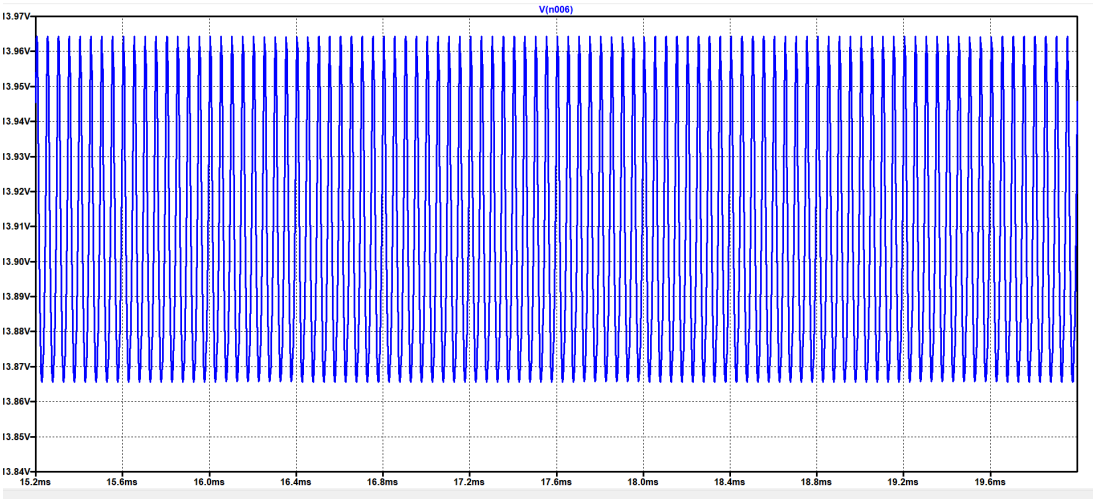

Figure 4: Simulated steady-state output voltage waveform (V_o), showing a stable average voltage of 13.91 V with a minimal peak-to-peak voltage ripple of 100 mV.

5. Firmware – PI Current Control

5.1. Timer1 PWM Configuration

Arduino Timer1 was configured in Fast PWM mode with ICR1 as the TOP value to generate the 20 kHz switching signal on pin 9:

$$f_{\text{PWM}} = \frac{F_{\text{CPU}}}{\text{ICR1} + 1} = \frac{16\,\text{MHz}}{799 + 1} = 20\,\text{kHz} \tag{14}$$

The starting OCR1A value for a 77.8 % non-inverting duty cycle:

$$\text{OCR1A} = D \times \text{ICR1} = 0.778 \times 799 \approx 621 \tag{15}$$

5.2. ACS712-05A Current Sensing

The ACS712-05A outputs a voltage proportional to current:

$$V_{\text{out}} = 2.5 + I \times 0.185 \quad [\text{V}] \tag{16}$$

$$I_{\text{mA}} = \frac{V_{\text{out}} - 2.5}{0.185} \times 1000 \tag{17}$$

At 500 mA per string (2 A total):

$$V_{\text{out}} = 2.5 + 2.0 \times 0.185 = 2.870\,\text{V} \tag{18}$$

5.3. PI Control Algorithm

```
//        Pin definitions


#define PWM_PIN    9
#define SENSOR_PIN A0

//        PI tuning parameters (tune on bench)


float Kp = 0.1f;
float Ki = 0.01f;

//        PI state variables


float integral  = 0.0f;
float pwm       = 621.0f;   // start at 77.8% D (non-inverted: OCR1A
    =621)
float setpoint  = 2000.0f;  // target = 2000 mA total (4 x 500 mA)

//        Timing


unsigned long lastTime  = 0;
```

```
int           loopCount = 0;

void setup() {
  // Timer1: Fast PWM, ICR1 as TOP, no prescaler -> 20 kHz on pin 9
  TCCR1A = _BV(COM1A1) | _BV(WGM11);
  TCCR1B = _BV(WGM13)  | _BV(WGM12) | _BV(CS10);
  ICR1   = 799;   // f = 16 MHz / (799+1) = 20 kHz
  OCR1A  = 621;   // D = 77.8% for IR2110

  Serial.begin(115200);
}

void loop() {
  //        Step 1: Read ACS712-05A on A0


  int   raw        = analogRead(SENSOR_PIN);
  float voltage    = (raw / 1023.0f) * 5.0f;
  float currentmA  = ((voltage - 2.5f) / 0.185f) * 1000.0f;

  //        Step 2: Error


  float error = setpoint - currentmA;

  //        Step 3: Timing


  unsigned long now = millis();
  float dt = (now - lastTime) / 1000.0f;
  lastTime = now;

  //        Step 4: PI with integral windup clamp

  integral += error * dt;
  integral  = constrain(integral, -1000.0f, 1000.0f);
  float PI_output = (Kp * error) + (Ki * integral);

  //        Step 5: Update PWM (non-inverting IR2110 logic)

  pwm = pwm + PI_output;
  pwm = constrain(pwm, 0.0f, 799.0f);
  OCR1A = (uint16_t)pwm;

  //        Step 6: Serial output every 10th loop

  loopCount++;
  if (loopCount >= 10) {
    Serial.print(F("I_mA: "));  Serial.print(currentmA, 1);
    Serial.print(F("  PWM: ")); Serial.print(pwm, 0);
    Serial.print(F("  Err: ")); Serial.println(error, 1);
    loopCount = 0;
  }

  delay(10);  // 10 ms control loop
}
```

Listing 1: Arduino PI Current Control – UV LED Buck Converter Driver

6. Germicidal Efficacy

6.1. UV Dose Calculation

The delivered UV dose depends on irradiance and exposure time:

$$H = E \times t \quad [\text{mJ/cm}^2] \tag{19}$$

At 265 nm, the action spectrum for *E. coli* DNA damage peaks, making the Klaran SA265 LEDs near-optimally matched to the germicidal requirement.

6.2. Dose-Response Data from Literature

Table 4: *E. coli* Inactivation at 265 nm

UV Dose (mJ/cm^2)	**Log Reduction**	**Kill Rate**
5–6	1-log	90 %
10–11	3-log	99.9 %
15–20	4-log	99.99 %
25+	5-log	99.999 %

A minimum optical output of 240 mW from the 8-LED array targets $15 - -20\,\text{mJ/cm}^2$ at the water surface, sufficient for the 99.99 % kill rate design goal.

7. PCB Design and Hardware Implementation

To transition the design from simulation to physical prototyping, a custom double-sided Printed Circuit Board (PCB) was schematic-captured and routed using KiCad EDA.

7.1. Complete KiCad Circuit Schematic

The KiCad schematic in Figure 5 integrates power conversion, auxiliary regulation, precision gate driving, and signal sampling:

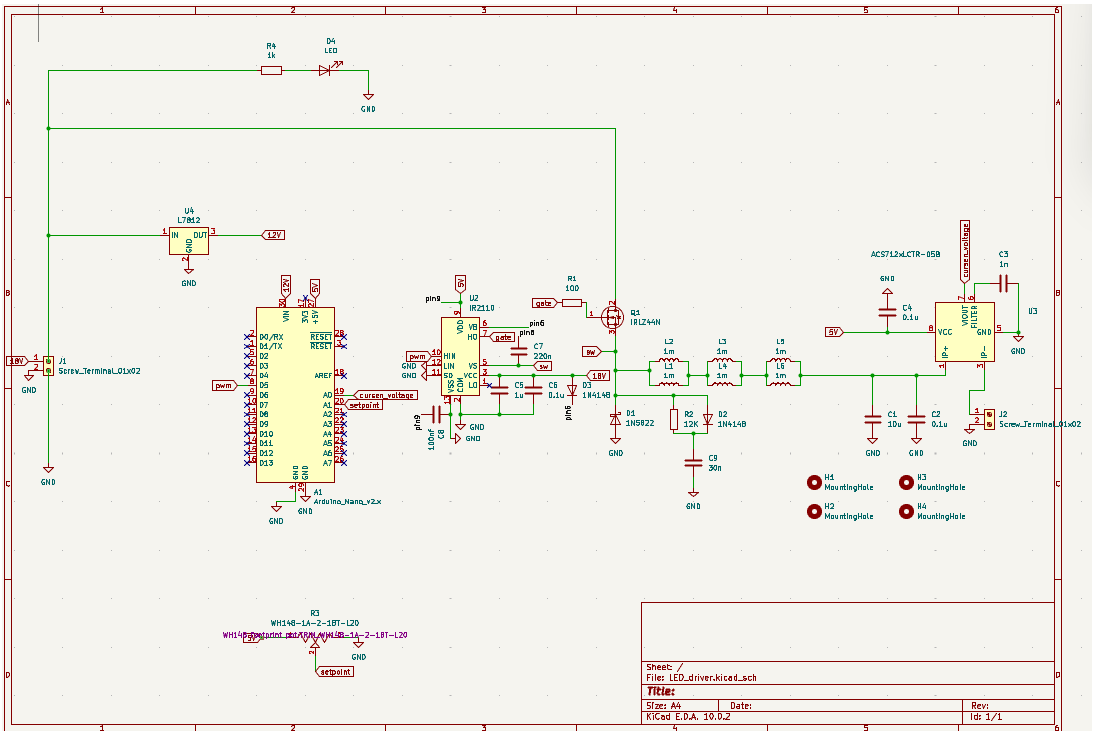

Figure 5: Complete circuit schematic generated in KiCad EDA showing the onboard L7812 regulator, IR2110 gate driver, ACS712 current sensor, microcontroller header, and inductor bank matrix.

7.2. Two-Layer PCB Layout Architecture

The physical layout (Figure 6) was optimized for low EMI, minimal parasitic loop inductance, and stable thermal dissipation:

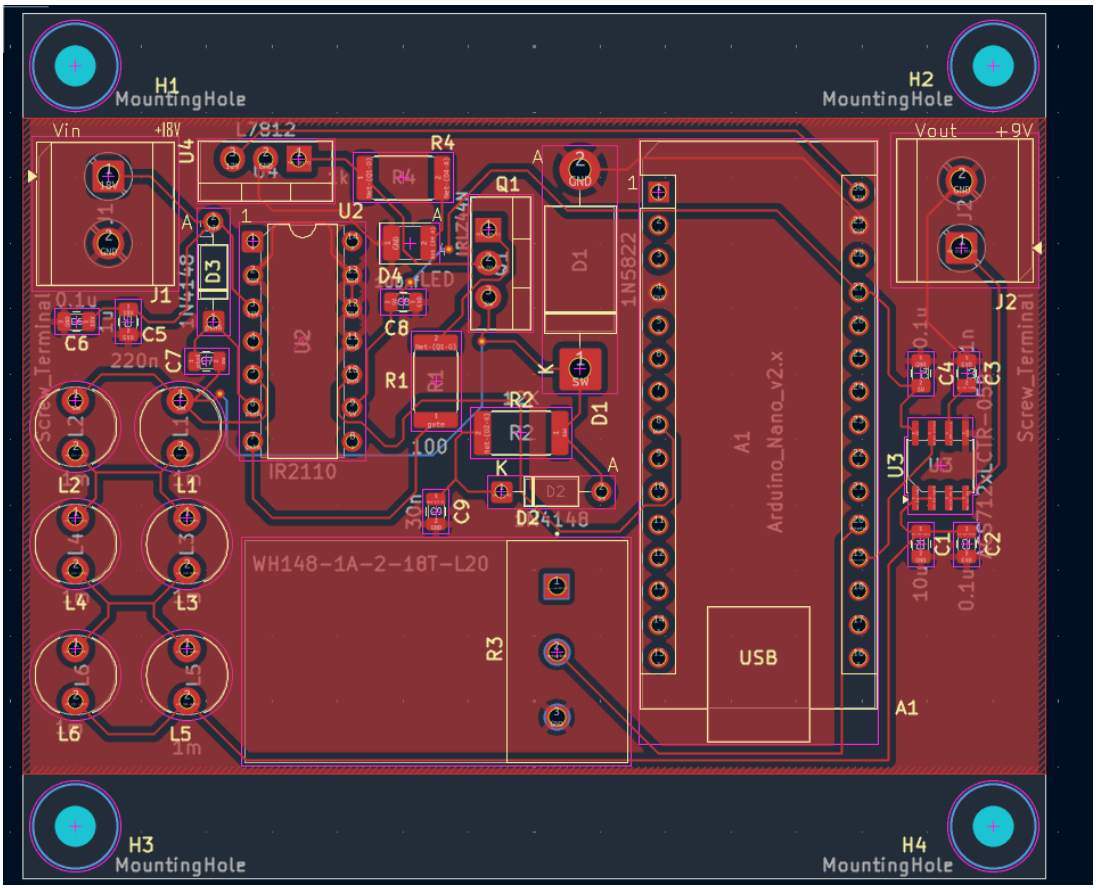

Figure 6: Two-layer PCB layout designed in KiCad, highlighting wide high-current power traces, thermal copper fills, isolated feedback signals, and standard M3 mounting holes.

7.3. Hardware Prototype Implementation

Prior to final PCB fabrication, the buck converter topology and closed-loop current feedback were verified on a breadboard setup (Figure 7). The hardware setup integrates the Arduino controller, switching power MOSFET, current feedback sensing circuitry, and passive components, allowing real-time calibration of the PI control parameters.

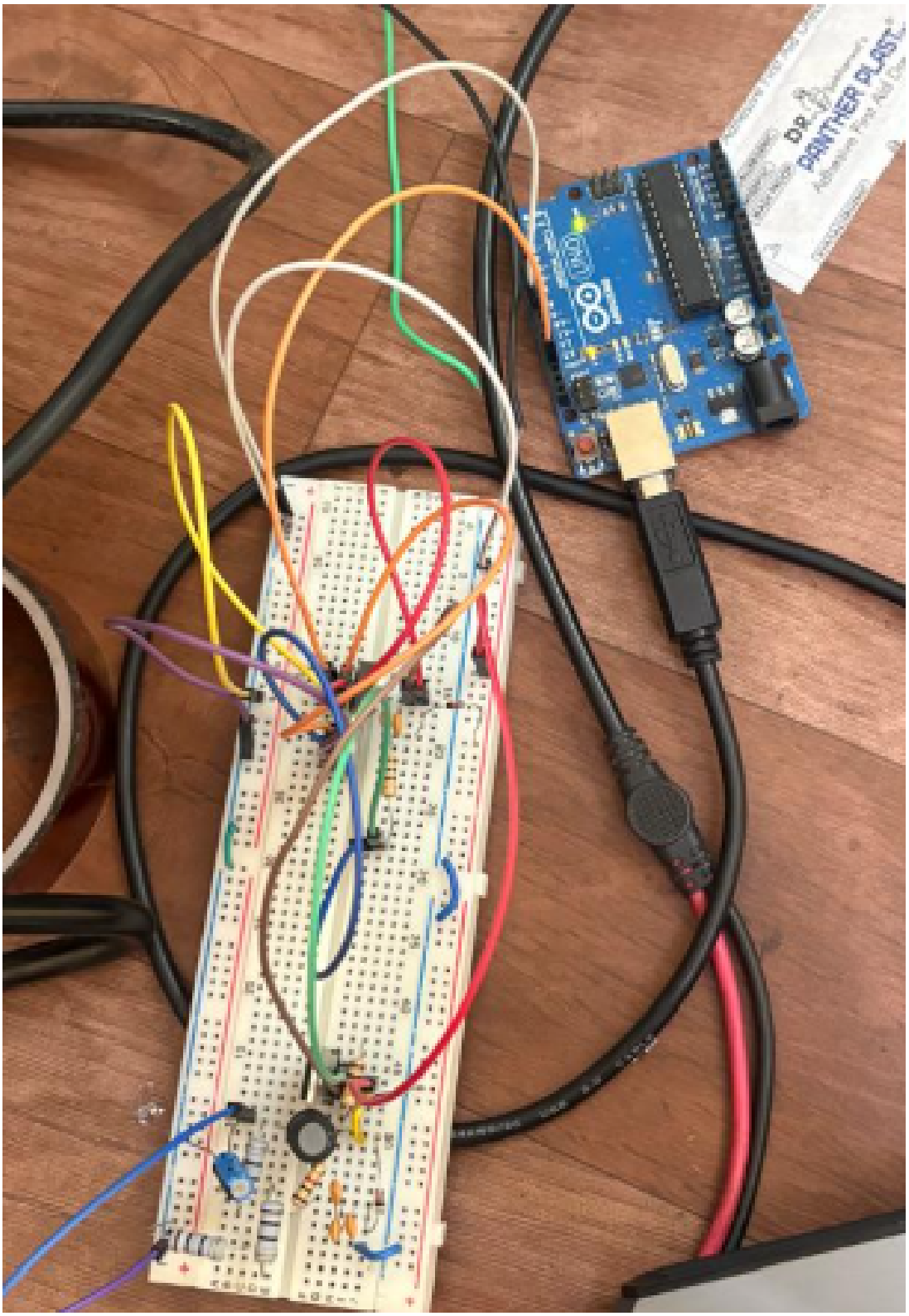

Figure 7: Breadboard hardware prototype setup of the UVC LED buck converter driver interfaced with an Arduino microcontroller during bench testing and parameter tuning.

8. Conclusion

A complete design flow was executed for a UVC LED buck converter driver targeting *E. coli* inactivation in water. The asynchronous converter (18 V $\rightarrow$ 14 V, 2 A) was designed, simulated in LTspice, and validated to operate in CCM with a 7.8 % inductor current ripple and $\Delta V_o = 0.10$ V output ripple.

Bench measurements on a DSO revealed key high-side switching challenges ($V_{\text{pk-pk}} = 32.8$ V ringing transients), justifying the implementation of an IR2110 driver IC and RCD snubber network. Closed-loop PI current control was implemented via Arduino firmware and ACS712 Hall-effect sensing. Finally, both breadboard prototyping and a custom 2-layer PCB layout were completed to ensure compact, reliable performance for water disinfection research.

References

[1] World Health Organization, *Guidelines for Drinking-water Quality*, 4th ed. Geneva: WHO Press, 2017.

[2] Crystal IS (Asahi Kasei), *Klaran SA Series UVC LEDs – Datasheet v2.0*, February 2024.

[3] J. Neuhaeusler and M. Xie, "Different Methods to Drive LEDs Using TPS63xxx Buck-Boost Converters," Texas Instruments Application Report SLVA419D, August 2021.

[4] M. Würtele *et al.*, "Application of GaN-based ultraviolet-C light emitting diodes – UV LEDs – for water disinfection," *Water Research*, vol. 45, pp. 1481–1489, 2011.